

2020 Practice

January 5, 2020

Problem S1: Plus

Time limit: 1.0 second(s)

Memory limit: 512 MB

Description

You are given two integers a and b , and you should output $a + b$.

Input Format

Read from standard input.

The only line contains two integers a and b , separated by one space.

Output Format

Write to the standard output.

Output only one integer $a + b$.

Sample Input 1

123 456

Sample Output 1

579

Subtasks

There are 5 subtasks. This task is for test purpose and constraints of all subtasks are the same.

Subtask 1 (10 points)

$0 \leq a, b \leq 100000$

Subtask 2 (15 points)

$0 \leq a, b \leq 100000$

Subtask 3 (20 points)

$$0 \leq a, b \leq 100000$$

Subtask 4 (25 points)

$$0 \leq a, b \leq 100000$$

Subtask 5 (30 points)

$$0 \leq a, b \leq 100000$$

Hint

You can use the code below to pass this task.

C

```
#include <stdio.h>
int main()
{
    int a, b;
    scanf("%d%d", &a, &b);
    printf("%d\n", a + b);
    return 0;
}
```

C++

```
#include <iostream>
int main()
{
    int a, b;
    std::cin >> a >> b;
```

```
std::cout << a + b << std::endl;  
return 0;  
}
```

Java

You must implement the main function in a public class **plus** (which must be the short name of the task), or you may get an error like Execution failed because the return code was nonzero.

```
import java.util.Scanner;  
public class plus {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        System.out.println(a + b);  
    }  
}
```

Problem S2: Circle

Time limit: 3.0 second(s)

Memory limit: 512 MB

Description

Given a circle consisting of n positive integers $a_0, a_1, \dots, a_{n-1}$ (a_i and a_{i+1} are adjacent for $0 \leq i < n-1$, while a_0 and a_{n-1} are also adjacent), each in the range $[1, 40]$. You can take two adjacent numbers with equal values and replace them by a single number of value one greater (e.g., you might replace two adjacent 7's with an 8). You can make the above moves as many times as you want. The goal is to maximize the value of the largest number present in the circle.

Now given $a_0, a_1, \dots, a_{n-1}$ and n , your task is to calculate the maximum value that can be obtained.

Input Format

Read from standard input.

The first line of the input contains one positive integer n .

While the second line of the input contains n positive integers $a_0, a_1, \dots, a_{n-1}$.

We have $n \leq 262,144$ and $a_0, a_1, \dots, a_{n-1} < 40$.

Output Format

Write to the standard output.

Output only one integer which represents the maximum value that can be obtained.

Sample Input 1

```
6
3 4 4 3 3 3
```

Sample Output 1

```
6
```

Sample Explanation 1

First we replace two pairs of adjacent 3's with two 4's and get totally four of 4's (note that a_0 and a_{n-1} are adjacent).

Then we replace two pairs of adjacent 4's with two 5's and finally get 6. Clearly there is no way to get an integer larger than 6.

Subtasks

Your submitted code will be judged in subtasks. You must pass all the cases in one subtask to get the score of this subtask.

Subtask	n	Score
1	≤ 32	10
2	≤ 512	30
3	$\leq 4,096$	40
4	$\leq 262,144$	20

Problem S3: Sqr

Time limit: 1.0 second(s)

Memory limit: 512 MB

Description

Given a perfect square number in binary representation, your task is to calculate its non-negative square root.

For example, we have $11001_{(2)} = 101_{(2)} \times 101_{(2)}$. If the input is **11011**, the output should be **101**.

Input Format

Read from standard input.

The input contains only one string with digits **0** and **1**, representing the perfect square number in binary.

Let n be its length, we promise $2 \leq n \leq 10^5$.

Output Format

Write to the standard output.

Output only one string with digits **0** and **1**, representing the non-negative square root of the given number in binary.

Sample Input 1

10010000

Sample Output 1

1100

Subtasks

There are 6 subtasks. The constraints for each subtask is shown as following:

Subtask	n	Score
1	≤ 10	10
2	≤ 50	10
3	≤ 60	10
4	≤ 300	30
5	$\leq 3,000$	30
6	$\leq 10^5$	10

Problem S4: Union

Time limit: 1.0 second(s)

Memory limit: 512 MB

Description

You are given an undirected connected graph of n vertices and $n - 1$ weighted edges (i.e. an unrooted tree with weighted edges). The vertices are numbered from 1 to n . The j -th edge connects vertices u_j and v_j , of weight w_j . The distance between two vertices is defined as the total weight of the edges on the unique path between them.

You need to maintain multiple sets, each containing several vertices of the graph. Initially there are n sets, where the i -th set only contains the i -th vertex. Then you need to handle $n - 1$ requests numbered from $n + 1$ to $2n - 1$. For request numbered i , you are given a_i and b_i , where $a_i < b_i < i$, and you need to assign the i -th set to be the union of the a_i -th and the b_i -th sets. It is guaranteed that for every set except set $2n - 1$, it is unioned (as parameter a or b) exactly once. Thus set $2n - 1$ must contain all the vertices.

Given an integer k , your task is to determine, for each set numbered from $n + 1$ to $2n - 1$, whether this set contains any 2 vertices between which the distance is at least k .

Input Format

Read from standard input.

The first line of the input contains 2 positive integers n and k .

The next $n - 1$ lines contain information of the edges, where the j -th line contains 3 integers u_j , v_j and w_j . It is guaranteed the input specifies an undirected connected graph.

The next $n - 1$ lines contain information about the requests, where the i -th line contains 2 integers a_{i+n} and b_{i+n} .

General constraints:

- $1 \leq n \leq 10^5$.
- $0 \leq w_i \leq 10^4$ for any i .
- $0 \leq k \leq 10^9$.

Output Format

Write to the standard output.

Output $n - 1$ lines of integer 0 or 1. For i -th line, output 1 if set $i + n$ contains 2 vertices between which the distance is at least k , or output 0 otherwise.

Sample Input 1

```
5 3
3 1 2
3 2 1
3 4 1
4 5 1
1 5
2 3
4 7
6 8
```

Sample Output 1

```
1
0
0
1
```

Sample Explanation 1

Set 6 is $\{1, 5\}$, and the distance between vertices 1 and 5 is 4, which is at least $k = 3$.

Subtasks

Your submitted code will be judged in subtasks. You must pass all the cases in one subtask to get the score of this subtask.

Subtask	n	Extra constraints	Score
1	≤ 2		5
2	≤ 200		20
3	$\leq 2,000$		20
4	$\leq 3 \times 10^4$	$u_j = j, v_j = j + 1$ for any j	20
5	$\leq 3 \times 10^4$		20
6	$\leq 10^5$		15